23CSE202 DATABASE MANAGEMENT SYSTEMS
S3 B.Tech CSE
Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus

PROJECT ABSTRACT

Group No : A8

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Project Title : Smart Digital Library Management System

Abstract :

Libraries in educational institutions often rely on manual or semi-digital methods to manage books, student records, borrowing transactions, and return schedules. These traditional approaches can lead to problems such as inaccurate record keeping, data redundancy, difficulty in tracking book availability, delayed updates, and inefficient management of overdue books and fines. As the number of books and library users increases, maintaining records manually becomes time-consuming and prone to errors. To address these challenges, this project proposes the development of a **Smart Digital Library Management System**, a database-driven application designed to automate and simplify library operations.

The primary objective of this project is to provide a centralized and efficient system for managing library resources while demonstrating the practical application of Database Management System (DBMS) concepts. The proposed system will maintain structured information about books, authors, publishers, categories, students, administrators, borrowing transactions, reservations, and fines using a relational database. The database will be designed with appropriate primary keys, foreign keys, constraints, and relationships to ensure data integrity, minimize redundancy, and support efficient data retrieval.

The intended users of the system are **library administrators** and **students**. Administrators will be able to manage book records, maintain student information, issue and return books, approve reservations, monitor overdue books, calculate fines automatically, and generate reports related to library activities. Students will be able to search for books, check their availability, reserve books that are currently unavailable, view their borrowing history, and track any pending fines.

The system will support major functionalities such as secure user login, book search with filtering options, book issue and return management, reservation handling, automatic fine calculation for overdue books, inventory management, and report generation. SQL operations such as Create, Read, Update, and Delete (CRUD), along with joins, aggregate functions, and relational queries, will be used extensively to manage and analyze library data.

The proposed Smart Digital Library Management System goes beyond basic data storage by integrating multiple related entities and supporting real-world library workflows. It demonstrates essential DBMS concepts including relational database design, normalization, referential integrity, and efficient query processing. By digitizing library operations, the system aims to improve efficiency, reduce manual effort, enhance data accuracy, and provide faster access to information for both administrators and students.